

INCOSE Systems Engineering Discovery Document - Simple Alarm Clock

1. Business or Mission Analysis

1.1 Problem Statement

People need a reliable and user-friendly way to wake up at a specific time or be reminded of events without manually monitoring the time.

1.2 Mission Objectives

Provide an alarm clock that:

- Accurately displays the current time.
- Allows the user to set a specific alarm time.
- Sounds a noticeable alarm when the current time matches the set alarm time.

1.3 Key Stakeholders (Business Level)

- **User:** The person who uses the clock to wake up or be reminded.
- **Manufacturer:** The company producing the alarm clock.

2. Stakeholder Needs and Requirements Definition

2.1 Use Cases / Operational Scenarios

- **View Time:** The user looks at the clock's display to know the current time.
- **Set Alarm:** The user inputs the desired wake-up time via the clock's interface.
- **Sound Alarm:** The clock detects that the current time matches the alarm time and produces a loud sound to alert the user.

2.2 Measures of Effectiveness (MOEs)

- **Time Accuracy:** The clock must maintain time accurately to within a minute per month.
- **Alarm Volume:** The alarm sound must be loud enough to wake a sleeping person (e.g., >70 dB).
- **Usability:** The interface for setting the alarm must be intuitive.

3. System Requirements Definition

3.1 Functional Requirements

- **REQ-FUNC-01 (Display Time):** The system shall display the current time continuously.
- **REQ-FUNC-02 (Set Alarm):** The system shall allow the user to set an alarm time.
- **REQ-FUNC-03 (Sound Alarm):** The system shall sound an alarm when the current time matches the alarm time.

3.2 Non-Functional / Quality Requirements

- **REQ-NF-01 (Volume):** The alarm shall be at least 70 dB to ensure it is heard.
- **REQ-NF-02 (Reliability):** The system shall trigger the alarm accurately when the set time arrives.

3.3 System Constraints

- **REQ-CON-01 (Power):** The system must operate on standard household power (or battery backup).

4. Architecture Definition

4.1 Logical Architecture / Subsystems

- **Timekeeping Subsystem:** Keeps track of the current time.
- **Display Subsystem:** Visually shows the time to the user.
- **Alarm Subsystem:** Compares current time with the set alarm time and triggers the alarm.
- **User Interface Subsystem:** Captures user inputs for setting time and alarm.
- **Sound Subsystem:** Produces the auditory alert.

4.2 Interfaces and Interactions

- The **User Interface Subsystem** provides user inputs to the **Alarm Subsystem** and **Timekeeping Subsystem**.
- The **Timekeeping Subsystem** sends the current time to the **Display Subsystem** and the **Alarm Subsystem**.
- The **Alarm Subsystem** sends a trigger signal to the **Sound Subsystem**.

4.3 System Behavior

- **Normal State:** Displaying the time.
- **Setting State:** User is setting the time or alarm.
- **Alarming State:** The alarm is sounding.

5. Design Definition

5.1 Physical Components

- **Microcontroller:** Acts as the central processor, managing timekeeping and logic.
- **LCD Display:** The physical screen showing the time.
- **Speaker:** The component producing the sound.
- **Buttons:** Physical interfaces for user input.
- **Power Supply:** Provides electrical power to components.

5.2 Allocation Matrix

- **Microcontroller** satisfies the **Timekeeping Subsystem** and **Alarm Subsystem**.
- **LCD Display** satisfies the **Display Subsystem**.
- **Speaker** satisfies the **Sound Subsystem**.
- **Buttons** satisfy the **User Interface Subsystem**.